

```
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity
```

```
# Customer profile = [Age, Income (₹ in thousands), Purchase Frequency]

customer1 = np.array([25, 50, 10])
customer2 = np.array([27, 55, 12])
```

```
print("Customer 1:", customer1)
print("Customer 2:", customer2)
```

```
Customer 1: [25 50 10]
Customer 2: [27 55 12]
```

```
euclidean = np.linalg.norm(customer1 - customer2)

print("Euclidean Distance:", euclidean)
```

```
Euclidean Distance: 5.744562646538029
```

```
manhattan = np.sum(np.abs(customer1 - customer2))

print("Manhattan Distance:", manhattan)
```

```
Manhattan Distance: 9
```

```
cosine = cosine_similarity([customer1], [customer2])

print("Cosine Similarity:", cosine[0][0])
```

```
Cosine Similarity: 0.9998398971353701
```

```
def edit_distance(s1, s2):
    m = len(s1)
    n = len(s2)

    dp = [[0]*(n+1) for _ in range(m+1)]

    for i in range(m+1):
        dp[i][0] = i

    for j in range(n+1):
        dp[0][j] = j

    for i in range(1, m+1):
        for j in range(1, n+1):
            if s1[i-1] == s2[j-1]:
                dp[i][j] = dp[i-1][j-1]
            else:
                dp[i][j] = 1 + min(dp[i-1][j],
                                   dp[i][j-1],
                                   dp[i-1][j-1])

    return dp[m][n]

name1 = "Rahul"
name2 = "RahulK"

print("Edit Distance:", edit_distance(name1, name2))
```

```
Edit Distance: 1
```

```
print("\nSimilarity Analysis")
print("Euclidean Distance :", euclidean)
print("Manhattan Distance :", manhattan)
print("Cosine Similarity :", cosine[0][0])
print("Edit Distance      :", edit_distance(name1, name2))
```

```
Similarity Analysis
Euclidean Distance : 5.744562646538029
Manhattan Distance : 9
```

Cosine Similarity : 0.9998398971353701
Edit Distance : 1

Start coding or generate with AI.